PROJECT DEVELOPMENT PHASE
Electricity Consumption Analysis

Project Development

The Project Development phase focuses on implementing the Electricity Consumption Analysis system using MySQL, Tableau Desktop, Tableau Public, and Flask. The development process involved collecting the dataset, storing it in a MySQL database, performing SQL analysis, creating interactive visualizations in Tableau, publishing the dashboard to Tableau Public, and integrating it into a Flask web application.

The project was developed systematically to ensure accurate analysis, meaningful visualization, and easy accessibility of electricity consumption insights.

Dataset Collection

The electricity consumption dataset was collected in CSV format. The dataset contains information such as:

- States
- Regions
- Latitude
- Longitude
- Dates
- Electricity Usage

The dataset was verified for consistency before importing it into the MySQL database.

Database Development

A MySQL database named **electricity** was created to store the dataset. The dataset was successfully imported into the **consumption** table.

SQL queries were executed to perform:

- Total Electricity Usage
- Average Electricity Usage
- Maximum Electricity Usage
- Minimum Electricity Usage
- State-wise Electricity Usage
- Region-wise Electricity Usage
- Top 10 Electricity Consumers

These queries helped generate meaningful analytical insights from the dataset.

Data Visualization

The MySQL database was connected to Tableau Desktop.

The following worksheets were created:

- State-wise Electricity Usage
- Region-wise Electricity Usage
- Monthly Electricity Usage Trend
- State-wise Electricity Usage Map
- Top 10 Electricity Consumers

Each worksheet was designed to provide interactive visual analysis of electricity consumption.

Dashboard Development

All worksheets were combined into a single interactive dashboard named:

Electricity Consumption Dashboard

The dashboard provides:

- State-wise comparison
- Region-wise comparison
- Monthly trend analysis
- Geographic visualization
- Top electricity-consuming states

Interactive filters were added to improve user experience and data exploration.

Story Development

A Tableau Story was created to present the project in a structured manner.
The story consists of:

1. Project Overview
2. Region-wise Electricity Usage
3. State-wise Electricity Usage
4. Monthly Electricity Usage Trend
5. State-wise Electricity Usage Map
6. Top 10 Electricity Consumers
7. Business Insights & Conclusion

The story explains the analysis and highlights the key findings.

Tableau Public Publishing

- After successful development of the dashboard and story, the project was published on Tableau Public.
- Publishing the dashboard allows users to access and interact with the visualizations online through a web browser.

Flask Web Application Development

- A Flask web application was developed using Python.
- The Tableau Public dashboard was embedded into the Flask application using HTML.
- The application allows users to access the dashboard through: http://127.0.0.1:5000
- This provides a simple web interface for viewing the dashboard.

GitHub Repository

The complete project was uploaded to GitHub.
The repository contains:

- Dataset
- SQL Queries
- Tableau Workbook
- Tableau Dashboard
- Story

- Flask Application
- Reports
- Screenshots
- README File
- Requirements File

This ensures proper project documentation and version control.

Development Workflow

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Electricity Dataset (CSV)
          |
          ▼
    MySQL Database
          |
          ▼
     SQL Analysis
          |
          ▼
   Tableau Desktop
          |
          ▼
Interactive Worksheets
          |
          ▼
      Dashboard
          |
          ▼
        Story
          |
          ▼
    Tableau Public
          |
          ▼
Flask Web Application
          |
          ▼
  GitHub Repository
```

Project Output

The final project successfully provides:

- Interactive electricity consumption dashboard.
- SQL-based data analysis.
- State-wise electricity usage.
- Region-wise electricity usage.
- Monthly electricity consumption trends.

- Geographic visualization using maps.
- Top 10 electricity-consuming states.
- Interactive Tableau Story.
- Web-based dashboard through Flask.
- Online dashboard through Tableau Public.